Aesthetic.Computer

A Planetary Laptop Orchestra

Jeffrey Alan Scudder

LACMA Art + Technology Lab · 2026

Aesthetic.Computer is a bare-metal creative computing system—custom hardware, a handmade programming language, and a social network—that reimagines the personal computer as a live musical instrument for art.

Project Description

Aesthetic.Computer (AC) is a creative computing platform built from first principles. It consists of three interlocking layers: a bare-metal operating system that boots directly into art software, a custom programming language called KidLisp for generative art, and a social network where anyone can publish and share interactive programs called "pieces."

The provocation is economic as much as aesthetic. Windows 10 end-of-life has stranded roughly 240 million x86_64 laptops; 62 million tonnes of e-waste are generated each year. Strip away the consumer operating system—notifications, app stores, surveillance—and those machines become a planetary population of half-built musical instruments waiting for a kernel.

AC Native is the kernel. A Linux boot that runs a custom C runtime as PID 1 on x86_64 UEFI laptops—no desktop, no window manager, no browser. It renders graphics through DRM without a compositor, reads input from raw evdev streams, and synthesizes audio sample-by-sample through ALSA at 192 kHz with 32-voice polyphony. Per-seat cost lands near $50, two orders of magnitude below Princeton's PLOrk laptop-orchestra model. A built-in `code` command drops into a native terminal running Anthropic's Claude Code—the only bare-metal creative OS we know of with an AI coding partner built in. The default piece is *notepat*, an 8,466-line polyphonic instrument with eight waveforms, room reverb, sample recording, and USB + UDP MIDI; twenty other pieces ship alongside it.

KidLisp is a minimal Lisp dialect designed specifically for generative art. With 118 built-in functions across 12 categories, it provides an accessible entry point for non-programmers while remaining expressive enough for complex compositions. Over 16,000 KidLisp programs have been written on the platform. KidLisp programs can be minted as on-chain "keeps" on Tezos, establishing provenance without requiring artists to understand blockchain infrastructure.

The Network ties it together. Aesthetic.Computer hosts 371 built-in pieces and 265 user-published pieces across 2,800+ registered handles. Every piece is URL-addressable and instantly shareable via QR code. The platform supports real-time multiplayer through WebSocket and UDP channels—people can draw, compose, and play together.

During the grant period, we propose to develop AC Native into a distributable creative instrument and public installation:

1. **Portable Instruments**—USB-bootable AC Native drives preloaded with curated pieces for visitors to take home and boot on their own laptops.
2. **KidLisp Workshops**—Hands-on sessions where participants write KidLisp programs that run on AC Native hardware in real time, experiencing the full loop from code to sound and image with no intermediary.
3. **Public Installation**—Multiple AC Native stations at LACMA where visitors encounter creative computing as a direct, embodied experience—more like sitting down at a piano than opening an app.
4. **Open Documentation**—Publish the complete build pipeline, hardware compatibility guide, and workshop curriculum so other artists and institutions can replicate the system.

This is not a product. It is an argument: the personal computer can still be a site of artistic invention—and the instrument is not yet finished being designed.

Figures

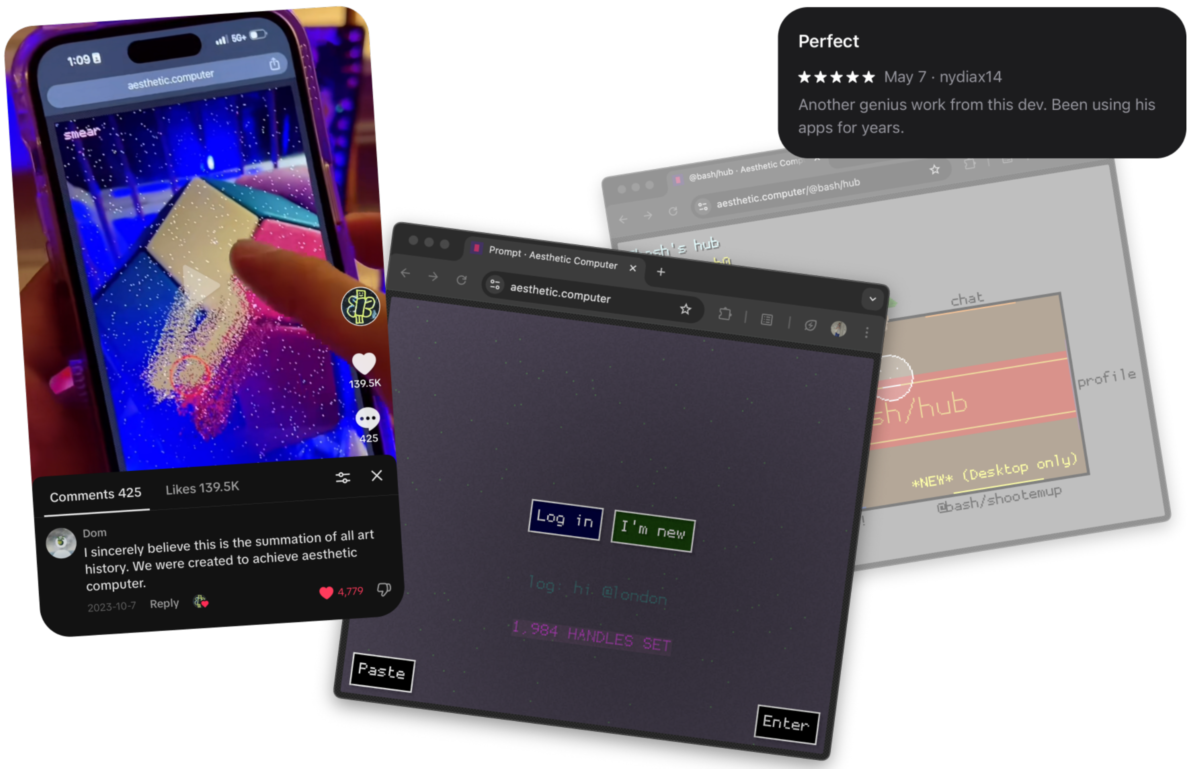


Fig. 1 — Aesthetic.Computer running on mobile and desktop. The platform hosts 600+ interactive pieces across 2,800 registered users. Social features include chat, multiplayer, and instant QR sharing.

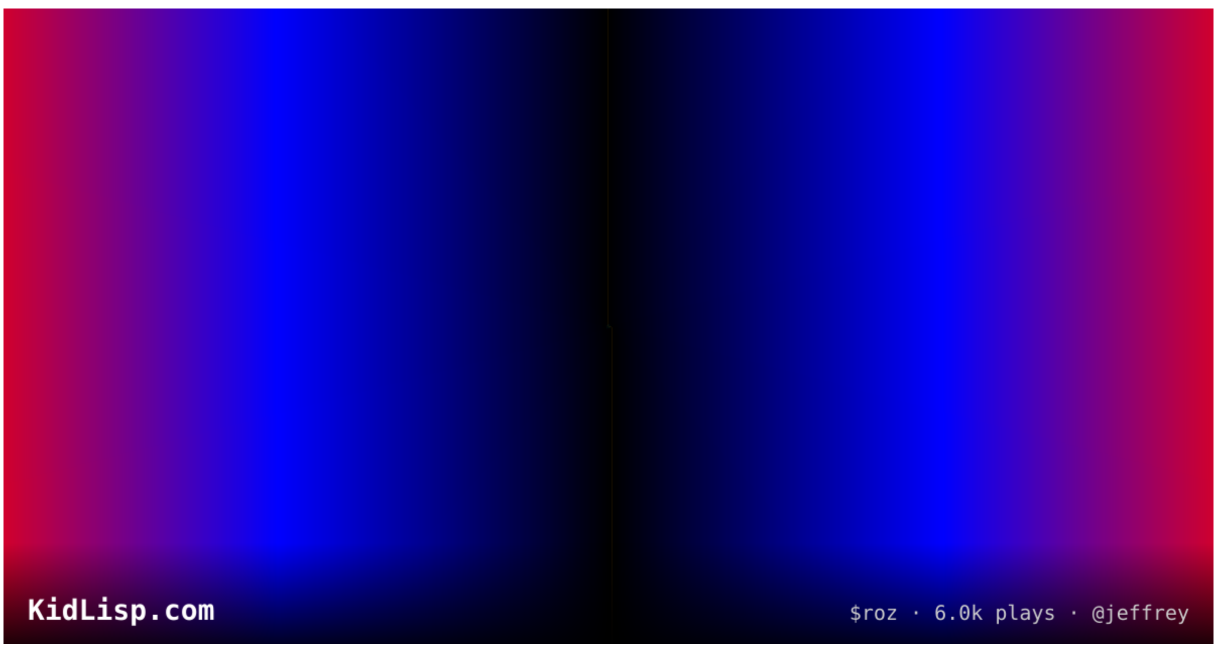


Fig. 2 — *$roz* by @jeffrey, a KidLisp generative piece with 6,000+ plays. KidLisp programs run

directly in the browser and on AC Native hardware. Over 16,000 programs have been written on the platform.

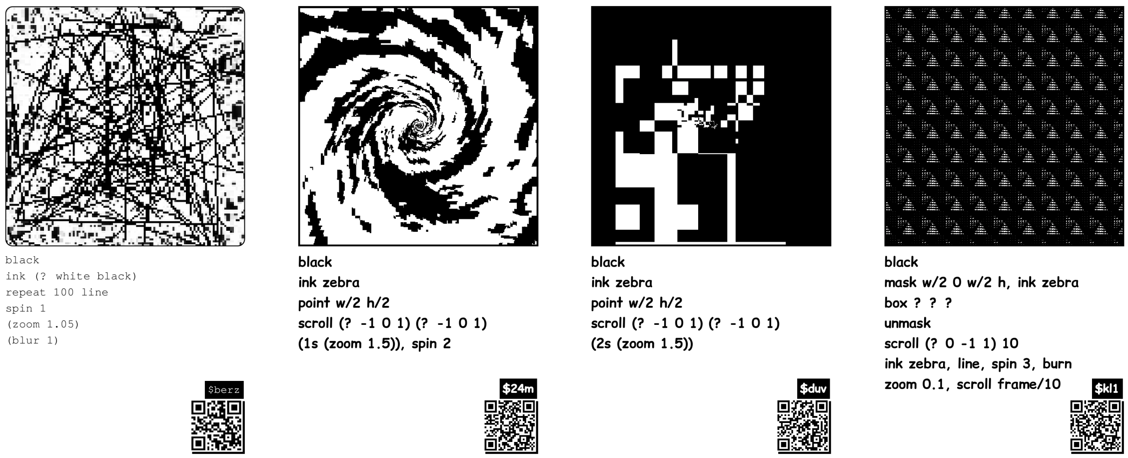
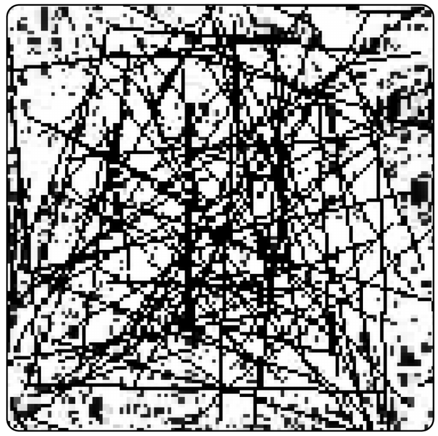


Fig. 3 — Four KidLisp pieces as printable cards: *$berz*, *$24m*, *$duv*, and *$kl1*. Each card is a self-contained program—the pixel image at top is the live output; the code underneath is the complete source. Printed cards were produced for Casey Reas & Lauren Lee McCarthy's "Social Software" course at UCLA (2026). QR codes link back to the live piece on Aesthetic.Computer.



```
black
ink (? white black)
repeat 100 line
spin 1
(zoom 1.05)
(blur 1)
```

$berz

Fig. 4 — *$berz* up close. Six lines of KidLisp produce a recursive wire-tangle that spins, zooms, and blurs each frame. No imports, no build step, no dependencies—the entire program is visible on the card. Cards are 2.75 $\times$ 4.75 inches, monochrome, screen-printable.

Fig. 5 — Target hardware: a Lenovo Yoga convertible laptop. AC Native boots from USB on x86_64 UEFI machines, turning commodity hardware into a dedicated creative instrument without replacing the factory OS.

Statements

Artistic Merit

Aesthetic.Computer treats the computer itself as an unfinished instrument—a site for ongoing artistic invention rather than a fixed consumer product. By building from bare metal (custom kernel, framebuffer rendering, sample-level audio synthesis), we recover the directness that early personal computing promised but commercial platforms abandoned. The work sits at the intersection of software art, instrument design, and language design: KidLisp is simultaneously a tool and a medium, and AC Native transforms commodity laptops into dedicated creative instruments. The artistic claim is that how we build computers is itself a creative act with cultural consequences.

Technology and Culture

Consumer operating systems have become attention-extraction machines—optimized for engagement metrics, not creative agency. AC Native offers a counter-model: a computer that boots directly into a single piece of art software and does nothing else. This is not nostalgia for early computing but a forward-looking argument that the personal computer's design is a cultural question, not a settled technical one. KidLisp extends this argument to programming itself—demonstrating that a language can be designed for artistic expression rather than industrial production. The 16,000+ programs written in KidLisp suggest this resonates beyond our own practice.

Public Engagement

We propose three forms of public engagement. First, hands-on KidLisp workshops at LACMA where participants write generative art programs that run on AC Native hardware—no prior coding experience required. Second, an installation of multiple AC Native stations where visitors experience creative computing as a direct, instrument-like interaction. Third, open "build days" where we assemble USB drives and document the process publicly, inviting visitors into the making of the system itself. All curricula, documentation, and software will be published openly for other artists and institutions to adopt.

Timeline

This proposal aligns to the Lab's new biennial calendar: a working-prototype milestone at the **2027 Symposium** and a completed public premiere at the **2028 Demo Day**. The cohort structure (3–5 recipients plus invitational projects) is well-suited to this work—AC Native's USB-bootable format means other cohort artists can bring their own pieces to the system, and we can cross-pollinate at Symposium without waiting for Demo Day. Two recent Lab projects sit directly in our neighborhood: Casey Reas's 2023 *METAVASARELY and An Empty Room* (generative systems) and Lauren Lee McCarthy's 2022 *Auto* (public/social interface design). Reas and McCarthy co-teach UCLA's Social Software course, which I am an Author in Residence in during this application year—their class is where the KidLisp cards shown in Fig. 3 were first printed and circulated.

Fall 2026 – Spring 2027	**Hardware & Curriculum**—Expand AC Native compatibility to 5+ laptop models; build multi-piece boot menu; design the 3-level KidLisp workshop curriculum; produce printed reference cards (extending the sosoft card template used at UCLA).
Spring – Summer 2027	**Pre-Symposium: First Workshops**—Run 2 pilot workshops at LACMA; assemble the first prototype multi-station installation; publish v0 of the open-source build guide.
Fall 2027	**2027 Symposium (works-in-development)**—Live AC Native demonstration: visitors (and cohort artists) boot the OS on their own laptops from a USB stick. Public KidLisp workshop in the museum. Talk / in-conversation on generative computing, situated alongside the 2023 cohort's work.
Winter 2027 – Summer 2028	**Full Installation + Extended Workshops**—Kiosk-mode hardening; 20+ bootable USB drives for visitor take-home; 4 additional workshops; complete documentation; translate workshop curriculum (English + Spanish, matching AC's existing translation pipeline).
Fall 2028	**2028 Demo Day (completed project)**—Public premiere of the multi-station AC Native installation in the LACMA galleries. Open-source v1.0 release with full build pipeline, hardware compatibility matrix, and workshop curriculum available for other institutions to adopt. Public programs introducing the system to teachers, museum educators, and other artists.

Budget

Artist fee (24 months, Fall 2026 – Fall 2028)	$22,000
Studio hardware (dev machines, displays)	$3,500
Installation laptops (5 × $400 refurbished)	$2,000
NuPhy analog keyboards for installation (5 × $120)	$600
USB drives, cables, peripherals	$500
Installation fabrication (furniture, mounts)	$2,500
Workshop materials (printed guides, KidLisp reference cards)	$1,200
2027 Symposium contribution (travel, cohort demo USB kit, on-site demo station)	$2,500
2028 Demo Day premiere (install setup, public-program support)	$3,000
Server + compute infrastructure (hosting, CDN, CI/CD)	$3,500
Documentation production (video, photography, translation)	$2,000
Contingency (10%)	$3,700
Total Requested	**$47,000**

Other funding: open-source sponsorship via GitHub Sponsors and Liberapay supplements ongoing infrastructure costs (hosting, domains, CI).

Curriculum Vitae

Jeffrey Alan Scudder · aesthetic.computer · Los Angeles, CA

Jeffrey Alan Scudder (b. 1989, Assonet, MA) is an artist based in Los Angeles, working across stretched canvas, custom software, and live performance. Yale School of Art MFA (2013). He is the creator of Aesthetic.Computer, Whistlegraph, and No Paint; his open-source tools *No Paint* (2020) and *notepat* (2024) each reached the front page of Hacker News. Work is held in the collections of KADIST (San Francisco) and SMK—National Gallery of Denmark. He is currently Author in Residence at UCLA, working with Casey Reas, and hosts biweekly NELA Computer Club demos at Plot.Place in Chinatown, Los Angeles.

Education

2013 Yale School of Art—MFA (Sculpture)

2011 Ringling College of Art + Design—BFA (Fine Art)

2010 AICAD New York Studio Program Residency

Residencies 2026 Author in Residence, UCLA (Casey Reas)
2019 Internet Archive Artist in Residence
2018 Gazell.io Art House Web Residency
2017 Schloss-Post Web Residencies (ZKM)

Teaching 2026 Author in Residence, UCLA (Casey Reas)
2024 UCLA DMA Summer: Interactivity
2019 Asst. Professor, Southern Oregon U.
2016 Visiting Professor, UCLA DMA
13–16 Adjunct Professor, Parsons

Selected Solo Exhibitions

2022 Ten Whistlegraphs, Feral File (w/ Whistlegraph)
2018 Radical Digital Painting: LA, The Newsstand Project & CAP
2017 Tumpin, left.gallery (Online)
2017 Imaginary Screenshots, Whitcher Projects, Los Angeles
2016 No Paint: Release One, Essex Flowers (Digital Commission)

Selected Group Exhibitions

2026 47th Venice Family Clinic Art Exhibition + Auction, Venice, CA
2025 Turbo Cheap (inaugural exhibition), Los Angeles
2019 Internet Archive Artist in Residency Exhibition
2018 OPEN CODES (Schloss Solitude Web Residencies), ZKM, Karlsruhe
2018 Radical Digital Painting w/ Julia Yerger, Johannes Vogt Gallery, NYC
2018 Make Pictures (curated), bitforms gallery, NYC
2015 P.S.1 Commencement, ALLGOLD @ MoMA PS1 Printshop, LIC, NY
2013 Apartment Show (hosted by Nouriel Roubini), NYC
2009 The New Easy, Artnews Projects, Berlin

Selected Lectures & Performances (65+ total)

2026 NELA Computer Club demos (biweekly), Plot.Place, Chinatown, LA
2022 The Longest Whistlegraph Ever (so far), New Museum, NYC
2020 New Dynamic Graphics: Whistlegraph Recital, Korea HCI (keynote)
2019 New Dynamic Graphics, India HCI 19 (keynote), Hyderabad
2019 RDP @ RAFLOST Festival, Iceland

2018 RDP, 35c3: Chaos Communication Congress, Leipzig
2018 Casey Reas & JAS in Conversation, bitforms gallery, NYC
2018 RDP @ 1st Annual HASH Award, ZKM, Karlsruhe
2018 RDP @ Pioneer Works, Brooklyn, NY
2018 European tour (15+ venues) w/ Goodiepal & Pals
2017 Northeast US Lecture Tour (Parsons, Harvard, Yale, RISD, Temple, Rutgers)
2017 RDP @ Cafe OTO, London & Neumeister Bar-Am, Berlin

Collections

KADIST Foundation, San Francisco
SMK—National Gallery of Denmark, Copenhagen

Selected Press & Writing

2024 *notepat*, Hacker News (front page)
2023 Whistlegraph—A new audience for generative art, *Dirt*
2023 Sex: The Whistlegraph Zine, *Sex Magazine*
2020 No Paint, Hacker News (front page)
2019 Drawing is the best videogame, *The Creative Independent*
2018 Radical Digital Painting & Political Rock, *Are.na Blog*
2018 *Harvard Advocate*, Winter 2018 (Noise)
2017 Artist Profile: Jeffrey Alan Scudder, *Rhizome*
2017 A Manifesto for Radical Digital Painting, *Schlosspost*
2017 Microsoft Paint's Influence on Artists Is Bigger Than You Might Think, *Artsy*

Software & Platforms

2021– Aesthetic.Computer—creative computing platform (aesthetic.computer)
2025– AC Native—bare-metal creative computing OS (USB-bootable Linux)
2024– KidLisp—programming language for generative art (16,000+ programs)
2024– *notepat*—polyphonic synthesizer instrument (8,466 lines)
2016– No Paint—radical paint program (nopaint.art)
2016– Whistlegraph—collaborative performance software (w/ Camille Klein & Alex Freundlich)

Appendix: LACMA 2026 Call for Proposals

Source: lacma.org/art/lab/grants · Retrieved April 2026

The Los Angeles County Museum of Art invites artists worldwide to apply for Art + Technology Lab grants. The initiative supports a "safe-to-fail rapid prototyping environment" where creators explore intersections of art, science, and emerging technology. The program emphasizes experimentation over finished products.

Key Details

Deadline	April 22, 2026, 11:59 PM PST
Notification	July 2026
Project Start	As early as fall 2026
Grant Amount	Up to $50,000 per project
Duration	Typically two years
Selected	3–5 via open call, plus up to 2 invitational

Eligibility

Artists and collectives worldwide may apply. Prior technological experience is not required. Applicants need not be Los Angeles–based.

Support Provided

Financial grants cover artist fees and direct costs including materials. Recipients also receive in-kind support: mentorship, coaching, and access to partner technologies from Hyundai Motor, Snap Inc., Anthropic, and academic institutions including MIT Media Lab Space Exploration Initiative and NASA Jet Propulsion Laboratory.

Evaluation Criteria

Projects must demonstrate:

- Artistic merit and artist-led vision
- Exploration of emerging technology
- Models or methods valuable to broader communities

- Public engagement opportunities during development

Application Requirements

- Project name and three descriptive words
- One-sentence description
- Full project description (500-word maximum)
- Artist biography / CV
- Artistic merit statement (100 words max)
- Discussion of technology–culture dialogue (100 words max)
- Public engagement plan (100 words max)
- Funding sources and conditions
- Detailed budget with milestones
- Up to 5 images / renderings (JPEG format)
- Video links (under 5 minutes)

Apply: lacma.submittable.com